

# Quantum Fault Tolerance Meets Toom's Contours

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## 1 Errors Clusters

**[COMMENT]** We should mention, that in any time that we call to  $H^{\otimes n}$ , even if its only for moving to the dual space, we immediately apply after the  $\phi$ -transpose mapping. The idea is that it will allow to us to ensure that the  $X$ -errors are always bounded by upper-right tingle and the  $Z$ -errors are always bounded by lower-left triangles.

**Definition 1.1** (**Size()**-Function). Consider a  $(d, d')$ -Toric complex  $\mathcal{G}$ , and let  $S$  be a subset of vertices,  $S \subset \mathcal{G}$ . We define the *size* of  $S$  to be:

$$\mathbf{Size}(S) := \sum_i^{d+1} \max_{v \in S} L_i(v)$$

Furthermore, we extend the notion to binary assignment over the  $i$ th faces of a complex  $z : \mathcal{F}^{(i)} \rightarrow \mathbb{F}_2$  by associating  $\mathbf{Size}(z)$  with the size of the vertex subset supporting  $z$ .

**Claim 1.2.** Let  $\mathbf{S} = \{S_1, S_2, \dots, S_k\}$  where  $S_i \subset \mathbb{R}^d$ , introduce an edge between  $S_i$  and  $S_j$  iff  $S_i \cap S_j \neq \emptyset$ . Assume that  $\mathbf{S}$  is connected. Then  $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$ .

*Proof.* It suffices to prove that if  $R_1 \cap R_2 \neq \emptyset$ , then  $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$ . Let  $a_{i,j} = \max_{v \in R_j} L_i(v)$ . Then for any  $w \in R_1 \cap R_2$  we have:

$$\begin{aligned} \mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\ &\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\ &= \mathbf{Size}(R_1) + \mathbf{Size}(R_2) \end{aligned}$$

□

**Definition 1.3.** error clusters  $(K_1, \dots, K_l, J_1, \dots, J_{l'})$ .

**Claim 1.4.** The following hold:

1.  $\phi \circ H^{\otimes n}$  sends  $X$ -type triangles to the same Size  $Z$ -type triangles, and vice versa.
2. **CX** and  $\varphi$  send  $X$ -type triangles to the same size  $X$ -type triangles. Similarly, they send  $Z$ -types triangles to the same size  $Z$ -type triangles.

*Proof.* Let  $z$  be a binary assignment over the  $d'$ -faces,  $z : \mathcal{F}^{(i)} \rightarrow \mathbb{F}_2$ , and let  $P$  be the Pauli product obtained by imposing a single-qubit Pauli  $X$  on every non-trivial coordinate of  $z$ . Then,  $\phi \circ HP$

$$L_g(\phi(z)) = \max_{v \in \phi(z)} v_g = \max_{v^{-1} \in z} -v_g$$

□

**Claim 1.5.** Let  $R_1, R_2$  be two registers, then any logical operator in  $\mathcal{O}$  sends error cluster to an error cluster as follows:

## 2 Toric Encoded Circuits

**Definition 2.1** (Troic Encoded Circuit). Let  $C$  be a quantum circuit. We say that  $C$  is *Toric encoded* if the following holds:

1. There is a Toric code  $\mathcal{C}$  such that the qubits of  $\mathcal{C}$  can be partitioned into registers  $R_1, \dots, R_k$ , and each of the registers is encoded by  $\mathcal{C}$ .
2. The gates in  $C$  with even time coordinates are sweep decoding gates.

For a set of faults  $\mathcal{E}$  the gates in  $C[\mathcal{E}]$  can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, \dots, D_1 P_1 \Lambda_1$$

**Definition 2.2** (Base graph  $G_o$ ). Let  $C$  be a quantum circuit, at depth  $d$ , with registers  $R_1, \dots, R_k$ , all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by  $\mathcal{G}_1, \dots, \mathcal{G}_k$  the Toric complexes that induce the codes. We define the base graph of  $C$ , denoted by  $G_o = (V_o, E_o)$  as follows:

1. The vertices  $V_o$  are tuples, where the first coordinates correspond to a vertex in the union over the vertices in  $\mathcal{G}_1, \dots, \mathcal{G}_k$ , and the second coordinate is associated with a time. Namely:

$$V_o := \bigsqcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges  $E_o$  are derived from the original edges in  $\mathcal{G}_1, \dots, \mathcal{G}_k$ , with the addition that any two vertices in preceding times  $t$  and  $t + 1$  that support qubits (faces)

such that  $C$  acts non-trivially on those qubits at time  $t$  are also connected by an edge. Namely:

$$E_o := \{ \{(v, t), (u, t')\} : \text{if} \\ \text{either there exists } i \text{ such } \{v, u\} \in \mathcal{G}_i \text{ and } t' = t, t + 1 \\ \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \} \}$$

**[COMMENT]** Here we should add that a vertex  $v$  belongs to location  $l$  if there is a qubit  $f$  that  $v$  belongs to its associated face.

**Definition 2.3** (Time Graph of  $C(\mathcal{E})$ ). For a quantum circuit  $C$  and set of faults  $\mathcal{E}$ , let us denote the base graph of  $C[\mathcal{E}]$  by  $G_o$ . For each time  $t$ , denote the deviation induced by  $\mathcal{E}$  at time  $t$  by  $\mathcal{D}_t$ , so  $\partial\mathcal{D}_t$  is the syndrome that the qubits in the deviation admit. We define the history graph  $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$  as follows:

1. The vertices  $V_{\mathcal{H}}$  are taken to be the vertices of  $G_o$ , namely tuples  $(v, t) \in V_o$ , that satisfy that  $v$  belongs to a face associated with a check, which, at time  $t$ , belongs to the  $\partial\mathcal{D}_t$ . Namely:

$$\{v_t : v_t = (v, t) \in V_o \text{ and there exists a check } f \in C \text{ such } v \in f \in \partial\mathcal{D}_t\}$$

2. The edges  $E_{\mathcal{H}}$  is partitioned into two types  $A_{\mathcal{H}}$  and  $F_{\mathcal{H}}$ , where  $A_{\mathcal{H}}$  are the edges in  $E_o$  restricted to  $V_{\mathcal{H}}$  and  $F_{\mathcal{H}}$  are defined to be the . Namely:

$$A_{\mathcal{H}} = \{e = \{u, v\} \in E_o : u, v \in V_{\mathcal{H}} \text{ and } T(v) = T(u) \pm 1\} \\ F_{\mathcal{H}} = \{e = \{u, v\} \in E_o : \text{exists } w \in V_{\mathcal{H}} \text{ such } \{v, w\}, \{u, w\} \in A_{\mathcal{H}}\}$$

By definition of the construction,  $G_{\mathcal{H}}$  is a time-graph.

**Definition 2.4** (Size Invariant Action). **[COMMENT]** Here we classify both transversal gates, and rotation on the Toric. An action  $g$  is said size invariant action if, for any assignment  $z$ , it holds that  $\mathbf{Size}(gz) = \mathbf{Size}(z)$ .

**Claim 2.5.** The sweep rule is Size  $\xi$ -shrinking.

*Proof.*

□

**[COMMENT]** Eventually, we might not used the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time =  $3t$ , in that we apply an invariant action so.